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RESEARCH

Non-fragile memory sampled-data control for exponential synchronization of uncertain complex-valued inertial neural networks with mixed delays

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Abstract This research focuses on the exponential synchronization of uncertain complex-valued inertial neural networks (CVINNs) with time-varying delays (TVDs) and distributed TVDs. Firstly, considering the fragility of controllers and transmission delays, a non-fragile memory sampled-data control (SDC) approach is employed to address the synchronization issue of CVINNs. Subsequently, to further reduce conservatism, the improved reciprocally convex inequality is extended to the complex domain. Furthermore, a Lyapunov-Krasovskii functional (LKF) based on the entire sampling period is designed, and combined with the improved complex-valued inequalities, sufficient conditions for the exponential synchronization of the addressed system are derived. Finally, a numerical simulation and an application example are developed, aiming to validate both theoretical efficacy and implementation feasibility.

Keywords Uncertain complex-valued inertial neural networks (CVINNs) · Mixed delays · Exponential synchronization · Non-fragile memory sampled-data control (SDC)

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1 Introduction

Neural networks (NNs), as key tools in machine learning and artificial intelligence, have garnered widespread utilization across domains such as pattern recognition, image processing, and natural language processing [1–3].Traditional NNs primarily handle real-valued data and achieve complex nonlinear mappings through multi-layer perceptron structures. However, with the increasing diversity of application demands, traditional models have been proven to be inadequate for some complex tasks, thereby which prompts the development of inertial neural networks (INNs). These networks introduce second-order inertial term to more accurately simulate the dynamic characteristics of biological neural systems, enabling richer behaviors when dealing with complex dynamical issues such as chaotic behavior and bifurcation phenomena [4,5]. The presence of inertial term allows the network to better capture transient behaviors in time series data, making them suitable for a broader range of dynamic system modeling and control problems [6–8]. Notably, complex-valued inertial neural networks (CVINNs) extend INNs by directly handling complex-valued inputs and outputs. Exploiting the representational prowess of complex domain, CVINNs adeptly manage high-dimen sional nonlinear challenges and are exceptionally suited for

sophisticated dynamic systems, thereby furnishing novel methodologies for addressing intricate problems [9–12].

Sampled-data control (SDC), based on discrete-time signal processing, is widely used in modern control systems. [13–18]. This technique entails sampling continuous system states at designated intervals, transforming continuous-time signals into discrete sequences, and subsequently employing these discrete data for the computation and execution of control laws. Compared to conventional continuous control methods, SDC offers numerous advantages, including diminished implementation costs, facilitation of system analysis and design, and enhanced convenience for data storage and remote control. Existing research has achieved advancements in the synchronization analysis of complex-valued neural networks (CVNNs) by utilizing sampled-data control [19–21]. Specifically, in [19], exponential stability of delayed complex-valued memristive neural networks was investigated, and a SDC was devised to achieve synchronization between master and slave systems. In [20], finite-time Mittag-Leffler synchronization criteria for fractional-order complex-valued reaction-diffusion neural networks were established via an impulsive and SDC. In [21], the asymptotic synchronization of delayed complex-valued stochastic switched neural networks under SDC was considered. Regrettably, there is currently a dearth of research concerning the synchronization for CVINNs with SDC, necessitating further exploration to bridge this gap.

In control theory, synchronization refers to the process in which interconnected systems gradually adjust their behaviors through mutual interactions or control inputs, ultimately achieving consistency in state evolution. It plays a significant role in applications such as coordinated control, secure communication, and complex network analysis. In contemporary control systems, time delays constitute an indispensable element that exerts a significant influence on system stability and dynamic behavior. Common types of delays encompass TVDs fluctuating over time [22,23] and distributed TVDs encapsulating the cumulative impact of past states on the current state across a defined interval [24,25]. In numerous practical systems, both types of delays may coexist, forming mixed delays. Analyzing systems with mixed delays is more challenging than single-delay systems, as it requires jointly considering time-varying and distributed delays, complicating stability analysis and control design. Consequently, substantial research efforts have been directed towards addressing synchronization issues in CVNNs with mixed delays [26–32]. However, there remains a paucity of studies concerning the synchronization of CVINNs with mixed delays, with relevant work appearing exclusively in [30–32], making it imperative to delve into the synchronization problem of CVINNs with mixed delays. How to reasonably design sampled-data controllers while taking into account the effects of mixed delays, in order to achieve efficient and stable operation of CVINNs under SDC, is a challenge worth considering.

Whether in the real or complex domain, system uncertainties, such as parameter uncertainties, external disturbances, represent a pivotal research focus within modern control theory. These uncertainties can precipitate performance degradation or even systemic instability. Consequently, comprehending and effectively managing these uncertainties is essential for the design of robust control systems. Although certain studies have tackled the synchronization challenges of uncertain CVNNs [33–35], the findings remain limited, and even fewer investigations have delved into the more intricate model of uncertain CVINNs [36]. To bolster system performance amidst such uncertainties, both non-fragile control and SDC assume critical roles. Non-fragile control focuses on controller uncertainties, ensuring robust performance under potential perturbations, whereas SDC efficiently mitigates data transmission requirements. However, guaranteeing the robustness of SDC in uncertain CVINNs remains an unresolved challenge, with scant existing literature addressing non-fragile SDC in this context. Furthermore, practical implementations necessitate consideration of transmission delays, which further complicate the analysis. Hence, there is an urgent imperative to develop a non-fragile memory SDC approach that accounts for both system and controller uncertainties as well as mixed delays, thereby providing a more comprehensive and reliable framework for CVINNs.

This paper aims to investigate the exponential synchronization problem of uncertain CVINNs with mixed delays through a non-fragile memory SDC scheme. The core contributions can be systematically characterized as:

(1) Unlike extant studies, this paper integrates the effects of parameter uncertainties, mixed delays,

and second-order inertial dynamics, thereby enhancing the practical relevance of the model and augmenting control performance and robustness.

(2) To mitigate conservatism, the refined reciprocal convex inequality proposed in [37] is extended to the complex domain, facilitating more precise estimation of the upper bound of time derivative of constructed Lyapunov-Krasovskii functional (LKF).

(3) This research tackles CVINNs in their original form, avoiding traditional techniques such as order reduction or decomposition into real and imaginary parts, thus preserving system's natural structure.

2 Preliminaries and Model Description

Notation. Let $\mathbb{R}^n$ ($\mathbb{C}^n$) denote the space of n-dimensional real (complex) vectors, and $\mathbb{R}^{n\times m}$ ($\mathbb{C}^{n\times m}$) denote the space of $n \times m$ real (complex) matrices. $*$ denotes the conjugate transpose of matrices. $\mathrm{Sym}\{P\} = P + P^*$. $\divideontimes$ indicates the conjugate transpose element in a Hermitian matrix. The notation $P > 0 (P < 0)$ represents that P is positive (negative) definite. I stands for the identity matrix with dimensions implied by context. $\lambda_{\min}(P)$ and $\lambda_{\max}(P)$ denote the minimum and maximum eigenvalues of matrix P, respectively. $|.|$ denotes the modulus for scalars, and $\|.\|$ denotes the 2-norm of a vector or a matrix.

Consider the following class of CVINNs subject to mixed delays and parameter uncertainties

$$\ddot{m}(t) = -\widetilde{A}(t)\dot{m}(t) - \widetilde{B}(t)m(t) + \widetilde{C}(t)f(m(t)) + \widetilde{D}(t)f\left(m^{\tau}(t)\right) + \widetilde{E}(t)\int_{t-l(t)}^{t} f(m(s))\,ds \tag{1}$$

where $m(t) \in \mathbb{C}^n$ is the state vector, $\ddot{m}(t)$ are the inertial term; $\widetilde{A}(t) \in \mathbb{R}^n$ and $\widetilde{B}(t) \in \mathbb{R}^n$ are positive-definite diagonal matrices; the connection weights $\widetilde{C}(t)$, $\widetilde{D}(t)$ and $\widetilde{E}(t)$ belong to the complex matrix space $\mathbb{C}^{n\times n}$; $f(m(t)) \in \mathbb{C}^n$ represents the complex-valued activation function; $m^{\tau}(t) = m(t-\tau(t))$. The time-varying delay $\tau(t)$ is bounded: $\tau_1 \leqslant \tau(t) \leqslant \tau_2$ and $\dot{\tau}(t) \leqslant d$; $l(t)$ stands for the distributed TVDs with $0 \leqslant l(t) \leqslant l$. Let $\bar{\tau} = \max\{\tau_2, l\}$, the initial conditions are $m(s) = \nu_1(s)$ and $\dot{m}(s) = \nu_2(s)$, $\forall s \in [-\bar{\tau}, 0]$. Weight matrices $\widetilde{\diamondsuit} = \diamondsuit + \Delta\diamondsuit(t)$, where $\diamondsuit = A, B, C, D, E$, $\diamondsuit$ are known matrices, and $\Delta\diamondsuit$ stand for uncertain parameters, which are assumed to be of the form

$$\begin{aligned}[\Delta A(t), \Delta B(t)] &= \mathfrak{M}_1 S(t)[\mathfrak{N}_1, \mathfrak{N}_2]\\ [\Delta C(t), \Delta D(t), \Delta E(t)] &= \mathfrak{M}_2 S(t)[\mathfrak{N}_3, \mathfrak{N}_4, \mathfrak{N}_5]\end{aligned} \tag{2}$$

where $S(t)$ is an unknown matrix satisfying $S(t)^* S(t) \leqslant I$. $\mathfrak{M}_1$, $\mathfrak{M}_2$, and $\mathfrak{N}_j$ ($j = 1, 2, ..., 5$) are known constant matrices.

Let $m(t) = \mathrm{col}\{m_1(t), m_2(t), \ldots, m_n(t)\}$ and $f(m(t)) = \mathrm{col}\{f_1(m_1(t)), f_2(m_2(t)), \ldots, f_n(m_n(t))\}$.

Assumption 1 For $f_q(m_q(t))$, $q = 1, 2, ..., n$, it is assumed that

$$|f_q(x) - f_q(y)| \leqslant \gamma_q |x - y|, \ \forall x, y \in \mathbb{C}$$

where $\gamma_q > 0$. Define $\Upsilon = \mathrm{diag}\{\gamma_1, \gamma_2, ..., \gamma_n\}$.

The corresponding response system follows

$$\ddot{\bar{m}}(t) = -\widetilde{A}(t)\dot{\bar{m}}(t) - \widetilde{B}(t)\bar{m}(t) + \widetilde{C}(t)f(\bar{m}(t)) + \widetilde{D}(t)f(\bar{m}^{\tau}(t)) + \widetilde{E}(t)\int_{t-l(t)}^{t} f(\bar{m}(s))ds + \Gamma U(t) \tag{3}$$

where $\bar{m}(t) \in \mathbb{C}^n$ is the state vector of response system, $U(t) \in \mathbb{C}^m$ represents the control input to be designed, and $\Gamma \in \mathbb{R}^{n\times n}$ is known matrix. The system (3) satisfies initial conditions: $\bar{m}(s) = \bar{\nu}_1(s)$, $\dot{\bar{m}}(s) = \bar{\nu}_2(s)$, $\forall s \in [-\bar{\tau}, 0]$.

The error system is derived by subtracting (3) from (1) as

$$\ddot{\hbar}(t) = -\widetilde{A}(t)\dot{\hbar}(t) - \widetilde{B}(t)\hbar(t) + \widetilde{C}(t)F(\hbar(t)) + \widetilde{D}(t)F\left(\hbar^{\tau}(t)\right) + \widetilde{E}(t)\int_{t-l(t)}^{t} F(\hbar(s))ds + \Gamma U(t) \tag{4}$$

where $\hbar(t) = \bar{m}(t) - m(t)$, $F(\hbar(t)) = f(\bar{m}(t)) - f(m(t))$, and $F(\hbar^{\tau}(t)) = f(\bar{m}^{\tau}(t)) - f(m^{\tau}(t))$.

The control signal is generated by a zero-order holder with the following hold time sequence: $0 = t_0 < t_1 ... < \lim_{k\to\infty} t_k = +\infty$. Define sampling interval is aperiodic and it satisfies $h_k = t_{k+1} - t_k \in (0, h]$, $\forall k \geqslant 0$.

The non-fragile sampled-data controller is designed in the following manner:

$$U(t) = (H + \Delta H)\hbar(t_k) + (H_\tau + \Delta H_\tau)\hbar^\eta(t_k) \quad (5)$$

where $\hbar^\eta(t_k) = \hbar(t_k - \eta)$, $t \in [t_k, t_{k+1})$, H and H_τ are the controller gain matrices with appropriate dimensions, η is a constant signal transmission delay, and ΔH and ΔH_τ represent the controller gain fluctuations satisfying

$$[\Delta H, \Delta H_\tau] = K\Theta[N, N_\tau] \quad (6)$$

where K, N, and N_τ are known constant matrices and $\Theta^*\Theta \leqslant I$.

Remark 1 In light of the presence of time delays within the system or signal transmission delays between the sampler and the controller, a memory sampled-data controller incorporating transmission delays η is designed to preserve the state information of the system continuously, thereby enhancing the efficacy of the closed-loop system performance. In addition, contrary to [34], which solely addresses time-varying uncertainties inherent in the system, our approach also encompasses time-varying uncertainties within the controller (6), leading to the development of a non-fragile control mechanism. Consequently, a non-fragile memory sampled-data controller is formulated for synchronization analysis, representing an inaugural investigation into this issue for uncertain CVINNs.

Substituting (5) and (4) yields to

$$\begin{aligned}\ddot{\hbar}(t) = &-\widetilde{A}(t)\dot{\hbar}(t) - \widetilde{B}(t)\hbar(t) + \widetilde{C}(t)F(\hbar(t)) \\ &+ \widetilde{D}(t)F\left(\hbar^\tau(t)\right) + \widetilde{E}(t)\int_{t-l(t)}^{t} F(\hbar(s))ds \\ &+ \Gamma(H + \Delta H)\hbar(t_k) \\ &+ \Gamma(H_\tau + \Delta H_\tau)\hbar^\eta(t_k) \end{aligned} \quad (7)$$

Definition 1 [38] The closed-loop system (7) is exponentially stable if there exist two positive constants a, b, such that

$$\|z(t)\| \leq b\exp(-at)\Big(\sup_{s\in[-r,0]}\{\|z(s)\|, \|\dot{z}(s)\|\}\Big)$$

where $r = \max\{\bar{\tau}, \eta\}$.

Lemma 1 *[39] For any continuous and differentiable function $\hbar : [a, b] \to \mathbb{C}^n$ and any positive define Hermitian matrix $\mathcal{P} \in \mathbb{C}^{n\times n}$, the following inequality holds*

$$\begin{aligned}&(b-a)\int_a^b \hbar^*(s)\mathcal{P}\hbar(s)ds \\ &\geqslant \begin{bmatrix}\int_a^b \hbar(s)ds \\ \xi\end{bmatrix}^* \begin{bmatrix}\mathcal{P} & 0 \\ 0 & 3\mathcal{P}\end{bmatrix}\begin{bmatrix}\int_a^b \hbar(s)ds \\ \xi\end{bmatrix}\end{aligned}$$

where $\xi = \int_a^b \hbar(s)ds - \frac{2}{b-a}\int_a^b\int_a^s \hbar(k)dkds$.

Lemma 2 *[39] For given positive integers p and q, a positive define Hermitian matrix $\mathcal{P} \in \mathbb{C}^{p\times p}$, a scalar $\nu \in (0, 1)$, matrices $\mathcal{W}_1$, $\mathcal{W}_2 \in \mathbb{C}^{p\times q}$, and any vector $\hbar \in \mathbb{C}^p$, if there exists a matrix $\mathcal{K} \in \mathbb{C}^{p\times p}$ satisfying $\begin{bmatrix}\mathcal{P} & \mathcal{K} \\ * & \mathcal{P}\end{bmatrix} > 0$, it holds that*

$$\begin{aligned}&\min_{\nu\in(0,1)}\left\{\frac{1}{\nu}\hbar^*\mathcal{W}_1\mathcal{P}\mathcal{W}_1\hbar + \frac{1}{1-\nu}\hbar^*\mathcal{W}_2\mathcal{P}\mathcal{W}_2\hbar\right\} \\ &\qquad\geqslant \begin{bmatrix}\mathcal{W}_1\hbar \\ \mathcal{W}_2\hbar\end{bmatrix}^* \begin{bmatrix}\mathcal{P} & \mathcal{K} \\ * & \mathcal{P}\end{bmatrix}\begin{bmatrix}\mathcal{W}_1\hbar \\ \mathcal{W}_2\hbar\end{bmatrix}\end{aligned}$$

Lemma 3 *[40] For any matrices $\mathcal{N}_1$ and $\mathcal{N}_2$ of appropriate dimensions and $v > 0$, then*

$$Sym\{\mathcal{N}_1\mathcal{N}_2\} \leqslant v^{-1}\mathcal{N}_1\mathcal{N}_1^* + v\mathcal{N}_2^*\mathcal{N}_2$$

Lemma 4 *[37] Let $\mathcal{Q}_1$, $\mathcal{Q}_2 \in \mathbb{R}^{n\times n}$ be positive definite matrices, ϖ_1, $\varpi_2 \in \mathbb{R}^n$, and a scalar $b \in (0, 1)$, then for any Y_1, $Y_2 \in \mathbb{R}^{n\times n}$, then*

$$\begin{aligned}&\frac{1}{b}\varpi_1^T\mathcal{Q}_1\varpi_1 + \frac{1}{1-b}\varpi_2^T\mathcal{Q}_2\varpi_2 \\ &\geq \varpi_1^T\big(\mathcal{Q}_1 + (1-b)(\mathcal{Q}_1 - Y_1\mathcal{Q}_2^{-1}Y_1^T)\big)\varpi_1 \\ &\quad + \varpi_2^T\big(\mathcal{Q}_2 + b(\mathcal{Q}_2 - Y_2^T\mathcal{Q}_1^{-1}Y_2)\big)\varpi_2 \\ &\quad + 2\varpi_1^T\big(bY_1 + (1-b)Y_2\big)\varpi_2\end{aligned} \quad (8)$$

Remark 2 In fact, further, the inequality (8) also holds in complex domain, that is, $\mathbb{R}^n$ and $\mathbb{R}^{n\times n}$ are extended to $\mathbb{C}^n$ and $\mathbb{C}^{n\times n}$, respectively. At this time, $\varpi_1 = \varpi_{11} + \varpi_{12}i$, $\varpi_2 = \varpi_{21} + \varpi_{22}i$, $\mathcal{Q}_1 = \mathcal{Q}_{11} + \mathcal{Q}_{12}i$, $\mathcal{Q}_2 = \mathcal{Q}_{21} + \mathcal{Q}_{22}i$, $Y_1 = Y_{11} + Y_{12}i$, and $Y_2 = Y_{21} + Y_{22}i$ with $\varpi_{11}, \varpi_{12}, \varpi_{21}, \varpi_{22} \in \mathbb{R}^n$, $\mathcal{Q}_{11}, \mathcal{Q}_{12}, \mathcal{Q}_{21}, \mathcal{Q}_{22}, Y_{11}, Y_{12}, Y_{21}, Y_{22} \in \mathbb{R}^{n\times n}$. Because $\mathcal{Q}_1$, $\mathcal{Q}_2$ are Hermitian matrices, then $\mathcal{Q}_{11} = \mathcal{Q}_{11}^T$, $\mathcal{Q}_{21} = \mathcal{Q}_{21}^T$,

$\mathcal{Q}_{12} = -\mathcal{Q}_{12}^T$, and $\mathcal{Q}_{22} = -\mathcal{Q}_{22}^T$. From (8), it follows that

$$\begin{aligned}
&\frac{1}{b}\varpi_1^*\mathcal{Q}_1\varpi_1 + \frac{1}{1-b}\varpi_2^*\mathcal{Q}_2\varpi_2 \\
&= \frac{1}{b}\begin{pmatrix}\varpi_{11}\\ \varpi_{12}\end{pmatrix}^T\begin{pmatrix}\mathcal{Q}_{11} & -\mathcal{Q}_{12}\\ \mathcal{Q}_{12} & \mathcal{Q}_{11}\end{pmatrix}\begin{pmatrix}\varpi_{11}\\ \varpi_{12}\end{pmatrix} \\
&+ \frac{1}{1-b}\begin{pmatrix}\varpi_{21}\\ \varpi_{22}\end{pmatrix}^T\begin{pmatrix}\mathcal{Q}_{21} & -\mathcal{Q}_{22}\\ \mathcal{Q}_{22} & \mathcal{Q}_{21}\end{pmatrix}\begin{pmatrix}\varpi_{21}\\ \varpi_{22}\end{pmatrix} \\
&\geq \begin{pmatrix}\varpi_{11}\\ \varpi_{12}\end{pmatrix}^T\left(\begin{pmatrix}\mathcal{Q}_{11} & -\mathcal{Q}_{12}\\ \mathcal{Q}_{12} & \mathcal{Q}_{11}\end{pmatrix} + (1-b)\left(\begin{pmatrix}\mathcal{Q}_{11} & -\mathcal{Q}_{12}\\ \mathcal{Q}_{12} & \mathcal{Q}_{11}\end{pmatrix} - \right.\right. \\
&\left.\left.\begin{pmatrix}Y_{11} & -Y_{12}\\ Y_{12} & Y_{11}\end{pmatrix}\begin{pmatrix}\mathcal{Q}_{21} & -\mathcal{Q}_{22}\\ \mathcal{Q}_{22} & \mathcal{Q}_{21}\end{pmatrix}^{-1}\begin{pmatrix}Y_{11} & -Y_{12}\\ Y_{12} & Y_{11}\end{pmatrix}^T\right)\right)\begin{pmatrix}\varpi_{11}\\ \varpi_{12}\end{pmatrix} \\
&+ \begin{pmatrix}\varpi_{21}\\ \varpi_{22}\end{pmatrix}^T\left(\begin{pmatrix}\mathcal{Q}_{21} & -\mathcal{Q}_{22}\\ \mathcal{Q}_{22} & \mathcal{Q}_{21}\end{pmatrix} + b\left(\begin{pmatrix}\mathcal{Q}_{21} & -\mathcal{Q}_{22}\\ \mathcal{Q}_{22} & \mathcal{Q}_{21}\end{pmatrix} - \right.\right. \\
&\left.\left.\begin{pmatrix}Y_{21} & -Y_{22}\\ Y_{22} & Y_{21}\end{pmatrix}^T\begin{pmatrix}\mathcal{Q}_{11} & -\mathcal{Q}_{12}\\ \mathcal{Q}_{12} & \mathcal{Q}_{11}\end{pmatrix}^{-1}\begin{pmatrix}Y_{21} & -Y_{22}\\ Y_{22} & Y_{21}\end{pmatrix}\right)\right)\begin{pmatrix}\varpi_{21}\\ \varpi_{22}\end{pmatrix} \\
&+ 2\begin{pmatrix}\varpi_{11}\\ \varpi_{12}\end{pmatrix}^T\left(b\begin{pmatrix}Y_{11} & -Y_{12}\\ Y_{12} & Y_{11}\end{pmatrix}\right. \\
&\left.+ (1-b)\begin{pmatrix}Y_{21} & -Y_{22}\\ Y_{22} & Y_{21}\end{pmatrix}\right)\begin{pmatrix}\varpi_{21}\\ \varpi_{22}\end{pmatrix} \\
&= \varpi_1^*(\mathcal{Q}_1 + (1-b)(\mathcal{Q}_1 - Y_1\mathcal{Q}_2^{-1}Y_1^*))\varpi_1 \\
&\quad + \varpi_2^*(\mathcal{Q}_2 + b(\mathcal{Q}_2 - Y_2^*\mathcal{Q}_1^{-1}Y_2))\varpi_2 \\
&\quad + 2\varpi_1^*(bY_1 + (1-b)Y_2)\varpi_2
\end{aligned}$$

Hence, the applicability scope of Lemma 4 has been extended. In the next section, we will apply the extended Lemma 4 to obtain less conservative results.

Lemma 5 *[41] If $h(t)$ and $x(t)$ are nonnegative continuous functions defined on $[0, T)$, and x_0 and T are positive constants such that the inequality*

$$x(t) \leqslant x_0(t) + \int_0^t h(s)\,x(s)ds, \quad t \in [0, T)$$

holds, then the following inequality holds:

$$x(t) \leqslant x_0 \exp\Big(\int_0^t h(s)ds\Big), \quad t \in [0, T)$$

Lemma 6 *[39] For any vector function $z : [a, b] \to \mathbb{C}^n$ and positive definite Hermitian matrix $P \in \mathbb{C}^{n\times n}$, the following linear matrix inequality (LMI) holds:*

$$\left(\int_a^b z(s)\,ds\right)^* P\left(\int_a^b z(s)\,ds\right) \leqslant (b-a)\int_a^b z^*(s)\,Pz(s)\,ds$$

where the integrals are assumed to be well-defined.

Lemma 7 *[42] 1) For scalers $\theta_p, \nu_p \in \mathbb{R}$, it yields*

$$\left(\sum_{p=1}^n \theta_p\nu_p\right)^2 \leqslant \sum_{p=1}^n \theta_p^2 \sum_{p=1}^n \nu_p^2,$$

where $p = 1, 2, ..., n$.

2) If $\varsigma(\varepsilon), \kappa(\varepsilon) : [a_1, a_2] \to \mathbb{R}$ are integrable and continuous functions on $[a_1, a_2]$, one can obtain

$$\left(\int_{a_1}^{a_2} \varsigma(\varepsilon)\,\kappa(\varepsilon)\,d\varepsilon\right)^2 \leqslant \int_{a_1}^{a_2} \varsigma^2(\varepsilon)\,d\varepsilon \int_{a_1}^{a_2} \kappa^2(\varepsilon)\,d\varepsilon.$$

3 Main Results

First, for convenience, the following notations are defined

$$\begin{aligned}
&u_\varsigma = \left[0_{n\times(\varsigma-1)n}\ I_{n\times n}\ 0_{n\times(22-\varsigma)n}\right],\ \varsigma = 1, 2, ..., 22 \\
&u_0 = 0_{n\times 22n},\ d_1(t) = t - t_k,\ d_2(t) = t_{k+1} - t, \\
&\tau_{12} = \tau_2 - \tau_1 \\
&v_1(t) = \frac{1}{d_1(t)}\int_{t_k}^t \hbar(s)ds,\ v_2(t) = \frac{1}{d_2(t)}\int_{t_k}^{t_{k+1}} \hbar(s)ds \\
&v_3(t) = \frac{1}{\eta}\int_{t-\eta}^t \hbar(s)ds,\ v_4(t) = \frac{1}{\tau_1}\int_{t-\tau_1}^t \hbar(s)ds \\
&\omega_1(t) = \mathrm{col}\{\omega_2(t), \hbar^\eta(t), \hbar^l(t)\} \\
&\omega_2(t) = \mathrm{col}\{\hbar(t), \dot{\hbar}(t)\} \\
&n_1(t) = \mathrm{col}\left\{\hbar(t) - \hbar(t_k), \hbar_\eta(t) - \hbar_\eta(t_k)\right\} \\
&n_2(t) = \mathrm{col}\left\{\hbar\left(t_{k+1}\right) - \hbar(t), \hbar_\eta\left(t_{k+1}\right) - \hbar_\eta(t)\right\} \\
&n_3(t) = \mathrm{col}\{d_2(t)n_1(t), d_1(t)n_2(t)\} \\
&n_4(t) = \mathrm{col}\{n_1(t), n_2(t)\} \\
&n_5(t) = \mathrm{col}\left\{\hbar(t_k), \hbar\left(t_{k+1}\right), \hbar^\eta(t_k), \hbar^\eta\left(t_{k+1}\right)\right\} \\
&n_6(t) = \mathrm{col}\{n_5(t), v_1(t), v_2(t)\} \\
&\phi_1(t) = \mathrm{col}\left\{F(\hbar(t)), F\left(\hbar^\tau(t)\right)\right\} \\
&\phi_2(t) = \mathrm{col}\left\{\hbar(t), \hbar^\tau(t)\right\} \\
&\zeta(t) = \mathrm{col}\{\omega_2(t), \ddot{\hbar}(t), \hbar^\tau(t), \hbar^{\tau_2}(t), \hbar^\eta(t), \dot{\hbar}^{\tau_2}(t), \\
&\qquad \dot{\hbar}^\eta(t), n_5(t), \phi_1(t), v_1(t), v_2(t), v_3(t), \hbar^{\tau_1}(t), \\
&\qquad v_4(t), \int_{t-l(t)}^t F(\hbar(s))ds, \hbar^l(t), \dot{\hbar}^l(t)\} \\
&\square^\diamond(t) = \square(t-\diamond),\ \square = \hbar, \dot{\hbar}, \omega_2,\ \diamond = \eta, l, \tau_1, \tau_2
\end{aligned}$$

Theorem 1 *Suppose Assumption 1 hold. For given positive scalars $h, a, \varepsilon_1, \varepsilon_2, \mu, r$, system (7) is exponentially stable, if there exist positive definite Hermitian matrices B, M_1, M_2, M_3, N_1, N_2, O_1, O_2, W_1, W_2, X, Hermitian matrix Y, arbitrary complex matrices S_1, S_2, S_3, Y_1, Y_2, G_1, G_2, G_3, G_4, $\mathcal{K}$; and real positive diagonal matrices $\emptyset_1$, $\emptyset_2$ such that*

$$\begin{bmatrix} M_1 & \mathcal{K} \\ * & M_1 \end{bmatrix} > 0 \tag{9}$$

$$\Omega_{11} - \mathcal{T}_1 \circledast_1^{-1} \mathcal{T}_1^* < 0 \tag{10}$$

$$\Omega_{11} + h\Omega_{22} - \mathcal{T}_2 \circledast_2^{-1} \mathcal{T}_2^* < 0 \tag{11}$$

$$\Omega_{11} + h\Omega_{33} - \mathcal{T}_3 \circledast_3^{-1} \mathcal{T}_3^* < 0 \tag{12}$$

where

$$\circledast_1 = [-\mu I, -\varepsilon_1 I, -\varepsilon_2 I, -\varepsilon_1 I, -\varepsilon_2 I, -\mu I]$$
$$\circledast_2 = [\circledast_1, -\exp(2ah)\,\mathcal{O}_2^{''}] \quad \circledast_3 = [\circledast_1, -\exp(2ah)\,\mathcal{O}_1^{''}]$$
$$\mathcal{T}_1 = [\mathbb{J}_0 \Gamma K, \mathbb{J}_0 \mathfrak{M}_1, \mathbb{J}_0 \mathfrak{M}_2, \varepsilon_1 (\mathfrak{N}_1 u_2 + \mathfrak{N}_2 u_1)^*, \varepsilon_2 (\mathfrak{N}_3 u_{13} + \mathfrak{N}_4 u_{14} + \mathfrak{N}_5 u_{20})^*, \mu (N u_9 + N_\tau u_{11})^*]$$
$$\mathcal{T}_2 = [\mathcal{T}_1, \Pi_1^* Y_1], \quad \mathcal{T}_3 = [\mathcal{T}_1, \Pi_2^* Y_2^*]$$
$$\begin{aligned}
\Omega_{11} = sym\Big\{ & \Phi_1^{1*} B \Phi_1^2 + 2a\Omega_1^{1*} S_3 \Omega_2^1 + \Omega_1^{2*} S_3 \Omega_2^1 \\
& + \Omega_1^{1*} S_3 \Omega_2^2 + \Omega_3^{5*} \left(S_1 \Omega_4^1 + S_2 \Omega_5 \right) \Big\} + 2a \Phi_1^{1*} B \Phi_1^1 \\
& + \tau_1^2 u_2^* M_3 u_2 + \Phi_2^{1*} M_2 \Phi_2^1 + \tau_{12}^2 u_2^* M_1 u_2 \\
& - e^{-2a\tau_2} \Phi_2^{2*} M_2 \Phi_2^2 - e^{-2a\tau_1} w_1^* M_3 w_1 \\
& - 3e^{-2a\tau_1} w_2^* M_3 w_2 + e^{-2a\tau_2} w_3^* Q w_3 \\
& + 2ah^2 \Omega_6^{1*} Y \Omega_6^1 + \Phi_2^{1*} N_1 \Phi_2^1 - e^{-2a\eta} \Phi_2^{3*} N_2 \Phi_2^3 \\
& + \eta^2 u_2^* N_2 u_2 - e^{-2a\eta} \varnothing_1^* N_2 \varnothing_1 + u_1^* X u_1 \\
& - 3e^{-2a\eta} \varnothing_2^* N_2 \varnothing_2 - e^{-2ah} \Pi_1^* O_1^{'} \Pi_1 \\
& - e^{-2ah} \Pi_2^* O_2^{'} \Pi_2 + l^2 u_{13}^* W_1 u_{13} - e^{-2al} u_{20}^* W_1 u_{20} \\
& + \Phi_2^{1*} W_2 \Phi_2^1 - e^{-2al} \Phi_2^{5*} W_2 \Phi_2^5 + \mathbb{J} \\
& - u_{15}^* \emptyset_1 u_{15} + u_1^* \Upsilon \emptyset_1 \Upsilon u_{15} - u_{16}^* \emptyset_2 u_{16} + u_4^* \Upsilon \emptyset_2 \Upsilon u_4
\end{aligned}$$
$$\begin{aligned}
\Omega_{22} = sym\Big\{ & 2a\Omega_3^{1*} \left(S_1 \Omega_4^1 + S_2 \Omega_5 \right) - \frac{1}{h_k} e^{-2ah} \Pi_1^* Y_2 \Pi_3 \\
& + \Omega_3^{3*} \left(S_1 \Omega_4^1 + S_2 \Omega_5 \right) + \Omega_3^{1*} S_1 \Omega_4^2 + \Omega_6^{2*} Y \Omega_6^1 \Big\} \\
& + \Omega_6^{1*} Y \Omega_6^1 + h_k u_2^* O_1 u_2 - \frac{1}{h_k} e^{-2ah} \Pi_1^* O_1^{'} \Pi_1
\end{aligned}$$
$$\begin{aligned}
\Omega_{33} = sym\Big\{ & 2a\Omega_3^{2*} \left(S_1 \Omega_4^1 + S_2 \Omega_5 \right) - \frac{1}{h_k} e^{-2ah} \Pi_1^* Y_1 \Pi_3 \\
& + \Omega_3^{4*} \left(S_1 \Omega_4^1 + S_2 \Omega_5 \right) + \Omega_3^{2*} S_1 \Omega_4^2 + \Omega_6^{3*} Y \Omega_6^1 \Big\} \\
& - \Omega_6^{1*} Y \Omega_6^1 + h_k u_2^* O_2 u_2 - \frac{1}{h_k} e^{-2ah} \Pi_2^* O_2^{'} \Pi_2
\end{aligned}$$
$$\begin{aligned}
\mathbb{J} = sym\{ & \mathbb{J}_0 (-u_3 - Au_2 - Bu_1 + Cu_{13} + Du_{14} \\
& + Eu_{20} + \Gamma H u_9 + \Gamma H_\tau u_{11}) \}
\end{aligned}$$
$$\mathbb{J}_0 = u_1^* G_1 + u_2^* G_2 + u_3^* G_3 + u_9^* G_4$$
$$\Phi_1^1 = col\left\{ \Phi_2^1, u_6, u_{21} \right\}, \ \Phi_2^1 = col\{u_1, u_2\}$$
$$\Phi_1^2 = col\{u_2, u_3, u_8, u_{22}\}, \ \Phi_2^2 = col\{u_5, u_7\}$$
$$\Phi_2^3 = col\{u_6, u_8\}, \ \Phi_2^5 = col\{u_{21}, u_{22}\}$$
$$\Omega_1^1 = col\{u_1 - u_9, u_6 - u_{11}\}, \ \Omega_1^2 = col\{u_2, u_8\}$$
$$\Omega_2^1 = col\{u_{10} - u_1, u_{12} - u_6\}, \ \Omega_2^2 = -\Omega_1^2$$
$$\Omega_4^1 = col\left\{ \Omega_1^1, \Omega_2^1 \right\}, \ \Omega_4^2 = col\left\{ \Omega_1^2, \Omega_2^2 \right\}$$
$$\Omega_5 = col\{u_9, u_{10}, u_{11}, u_{12}\}, \ \Omega_6^1 = col\{\Omega_5, u_{15}, u_{16}\}$$
$$\hat{u}_0 = col\{u_0, u_0, u_0\}, \ \Omega_6^2 = col\left\{ \hat{u}_0, \hat{u}_0, u_1 - u_{15,} u_0 \right\}$$
$$\Omega_6^3 = col\left\{ \hat{u}_0, \hat{u}_0, u_0, u_{16} - u_{1,} \right\}, \ \Omega_3^1 = col\left\{ \Omega_1^1, \hat{u}_0 \right\}$$
$$\Omega_3^2 = col\left\{ \hat{u}_0, \Omega_2^1 \right\}, \ \Omega_3^3 = col\left\{ \Omega_1^2, \hat{u}_0 \right\}$$
$$\Omega_3^4 = col\left\{ \hat{u}_0, \Omega_2^2 \right\}, \ \Omega_3^5 = col\left\{ -\Omega_1^1, \Omega_2^1 \right\}$$
$$\varnothing_1 = u_1 - u_6, \varnothing_2 = u_1 + u_6 - 2u_{17}$$
$$\Pi_1 = col\{u_1 - u_9, u_1 + u_9 - 2u_{15}\}$$
$$\Pi_2 = col\{u_{10} - u_1, u_1 + u_{10} - 2u_{16}\}$$
$$\Pi_3 = col\{u_{10} - u_1, 3(u_1 + u_{10} - 2u_{16})\}$$
$$\Theta_1 = u_1 - u_{18}, \Theta_2 = u_1 + u_{18} - 2u_{19}$$
$$\Theta_3 = col\{u_4, u_5, u_{18}\}$$

Proof Construct the following LKF

$$V(t) = \sum_{j=1}^{7} V_j(t), \ t \in [t_k, t_{k+1}) \tag{13}$$

where

$$V_1(t) = e^{2at} \omega_1(t) B \omega_1(t)$$
$$\begin{aligned}
V_2(t) = & 2e^{2at} n_3^*(t) (S_1 n_4(t) + S_2 n_5(t)) \\
& + 2e^{2at} n_1^*(t) S_3 n_2(t)
\end{aligned}$$
$$V_3(t) = d_1(t) d_2(t) e^{2at} n_6^*(t) Y n_6(t)$$
$$\begin{aligned}
V_4(t) = & \tau_{12} \int_{-\tau_2}^{-\tau_1} \int_{t+\theta}^{t} e^{2as} \dot{\hbar}^*(s) M_1 \dot{\hbar}(s) ds d\theta \\
& + \int_{t-\tau_2}^{t} e^{2as} \omega_2^*(s) M_2 \omega_2(s) ds \\
& + \tau_1 \int_{-\tau_1}^{0} \int_{t+\theta}^{t} e^{2as} \dot{\hbar}^*(s) M_3 \dot{\hbar}(s) ds d\theta
\end{aligned}$$
$$V_5(t) = \int_{t-\eta}^{t} e^{2as} \omega_2^*(s) N_1 \omega_2(s) ds$$

$$+\eta\int_{-\eta}^{0}\int_{t+\theta}^{t}e^{2as}\dot{\hbar}^{*}(s)\,N_2\dot{\hbar}(s)dsd\theta$$

$$V_6(t)=d_2(t)h_k\int_{t_k}^{t}e^{2as}\dot{\hbar}^{*}(s)\,O_1\dot{\hbar}(s)ds$$
$$-d_1(t)h_k\int_{t}^{t_{k+1}}e^{2as}\dot{\hbar}^{*}(s)\,O_2\dot{\hbar}(s)ds$$
$$+\int_{t_k}^{t}e^{2as}\hbar^{*}(s)\,X\hbar(s)ds$$

$$V_7(t)=l\int_{-l}^{0}\int_{t+\theta}^{t}e^{2as}F^{*}(\hbar(s))\,W_1F(\hbar(s))dsd\theta$$
$$+\int_{t-l}^{t}e^{2as}\omega_2^{*}(s)\,W_2\omega_2(s)ds$$

Next, we will calculate the time derivative of $V(t)$. For $t\in[t_k,t_{k+1})$, first,

$$\dot{V}_1(t)=e^{2at}\Big(2a\omega_1^{*}(t)\,B\omega_1(t)+2\omega_1^{*}(t)\,B\dot{\omega}_1(t)\Big) \tag{14}$$

$$\dot{V}_2(t)=e^{2at}\Big(4an_1^{*}(t)S_3n_2(t)+2\dot{n}_1^{*}(t)S_3n_2(t)$$
$$+2n_1^{*}(t)\,S_3\dot{n}_2(t)+4an_3^{*}(t)\,(S_1n_4(t)+S_2n_5(t))$$
$$+2\dot{n}_3^{*}(t)\,(S_1n_4(t)+S_2n_5(t))+2n_3^{*}(t)\,S_1\dot{n}_4(t)\Big) \tag{15}$$

$$\dot{V}_3(t)=e^{2at}\Big(2ad_1(t)d_2(t)n_6^{*}(t)\,Yn_6(t)$$
$$+2d_1(t)d_2(t)\dot{n}_6^{*}(t)\,Yn_6(t)+d_2(t)n_6^{*}(t)\,Yn_6(t)$$
$$-d_1(t)n_6^{*}(t)\,Yn_6(t)\Big) \tag{16}$$

Second, for $\dot{V}_4(t)$, by applying Jensen inequality in the complex number field [39], Lemmas 1 and 2, one can deduce that

$$\dot{V}_4(t)\leqslant e^{2at}\Big(\tau_{12}^2\dot{\hbar}^{*}(t)\,M_1\dot{\hbar}(t)+\tau_1^2\dot{\hbar}^{*}(t)\,M_3\dot{\hbar}(t)$$
$$+\omega_2^{*}(t)\,M_2\omega_2(t)-e^{-2a\tau_2}\omega_2^{\tau_2*}(t)\,M_2\omega_2^{\tau_2}(t)$$
$$-\tau_{12}e^{-2a\tau_2}\int_{t-\tau_2}^{t-\tau_1}\dot{\hbar}^{*}(s)\,M_1\dot{\hbar}(s)ds$$
$$-\tau_1e^{-2a\tau_1}\int_{t-\tau_1}^{t}\dot{\hbar}^{*}(s)\,M_3\dot{\hbar}(s)ds\Big)$$
$$\leqslant e^{2at}\Big(\tau_{12}^2\dot{\hbar}^{*}(t)\,M_1\dot{\hbar}(t)+\tau_1^2\dot{\hbar}^{*}(t)\,M_3\dot{\hbar}(t)$$
$$+\omega_2^{*}(t)\,M_2\omega_2(t)-e^{-2a\tau_2}\omega_2^{\tau_2*}(t)\,M_2\omega_2^{\tau_2}(t)$$
$$+e_1^{*}Qe_1-\tau_1e^{-2a\tau_1}\int_{t-\tau_1}^{t}\dot{\hbar}^{*}(s)\,M_3\dot{\hbar}(s)ds\Big) \tag{17}$$

where

$$e_1=\mathrm{col}\left\{\hbar^{\tau}(t),\hbar^{\tau_2}(t),\hbar^{\tau_1}(t)\right\}$$
$$Q=\begin{bmatrix}-2M_1+\mathrm{sym}\{\mathcal{K}\} & M_1-\mathcal{K} & M_1-\mathcal{K}^{*}\\ \ast & -M_1 & \mathcal{K}^{*}\\ \ast & \ast & -M_1\end{bmatrix}$$

For $\dot{V}_5(t)$, according to Lemma 1, we have

$$\dot{V}_5(t)\leqslant e^{2at}\Big(\omega_2^{*}(t)\,N_1\omega_2(t)-e^{-2a\eta}\omega_2^{\eta*}(t)\,N_1\omega_2^{\eta}(t)$$
$$+\eta^2\dot{\hbar}^{*}(t)\,N_2\dot{\hbar}(t)-\eta e^{-2a\eta}\int_{t-\eta}^{t}\dot{\hbar}^{*}(s)\,N_2\dot{\hbar}(s)ds\Big)$$
$$\leqslant e^{2at}\Big(\omega_2^{*}(t)\,N_1\omega_2(t)-e^{-2a\eta}\omega_2^{\eta*}(t)\,N_1\omega_2^{\eta}(t)$$
$$+\eta^2\dot{\hbar}^{*}(t)\,N_2\dot{\hbar}(t)+e^{-2a\eta}\left(-l_1^{*}N_2l_1-3l_2^{*}N_2l_2\right)\Big) \tag{18}$$

where $l_1=\hbar(t)-\hbar^{\eta}(t)$, $l_2=\hbar(t)+\hbar^{\eta}(t)-2v_3(t)$.

For $\dot{V}_6(t)$, according to Lemmas 1 and 4 and Remark 2, we have

$$\dot{V}_6(t)\leqslant e^{2at}\Big(d_2(t)h_k\dot{\hbar}^{*}(t)\,O_1\dot{\hbar}(t)+d_1(t)h_k\dot{\hbar}^{*}(t)\,O_2\dot{\hbar}(t)$$
$$+\hbar^{*}(t)\,X\hbar(t)-e^{-2ah}h_k\int_{t_k}^{t}\dot{\hbar}^{*}(s)\,O_1\dot{\hbar}(s)ds$$
$$-e^{-2ah}h_k\int_{t}^{t_{k+1}}\dot{\hbar}^{*}(s)\,O_2\dot{\hbar}(s)ds)$$
$$\leqslant e^{2at}\Big(d_2(t)h_k\dot{\hbar}^{*}(t)\,O_1\dot{\hbar}(t)+\hbar^{*}(t)\,X\hbar(t)$$
$$+d_1(t)h_k\dot{\hbar}^{*}(t)\,O_2\dot{\hbar}(t)$$
$$-e^{-2ah}\Big(p_1^{*}\Big(O_1'+\frac{d_2(t)}{h_k}\Big(O_1'-Y_1O_2^{''-1}Y_1^{*}\Big)\Big)p_1$$
$$+p_2^{*}\Big(O_2'+\frac{d_1(t)}{h_k}\Big(O_2'-Y_2^{*}O_1^{''-1}Y_2\Big)\Big)p_2$$
$$+\mathrm{sym}\Big\{p_1^{*}\Big(\frac{d_1(t)}{h_k}Y_1+\frac{d_2(t)}{h_k}Y_2\Big)p_3\Big\}\Big)\Big) \tag{19}$$

where

$$p_1=\mathrm{col}\{\hbar(t)-\hbar(t_k),\hbar(t)+\hbar(t_k)-2v_1(t)\}$$
$$p_2=\mathrm{col}\left\{\hbar(t_{k+1})-\hbar(t),\hbar(t)+\hbar(t_{k+1})-2v_2(t)\right\}$$
$$p_3=\mathrm{col}\left\{\hbar(t_{k+1})-\hbar(t),3\left(\hbar(t)+\hbar(t_{k+1})-2v_2(t)\right)\right\}$$
$$O_1'=\mathrm{diag}\{O_1,3O_1\},\ O_2'=\mathrm{diag}\{O_2,3O_2\}$$
$$O_1''=\mathrm{diag}\left\{O_1,\frac{1}{3}O_1\right\},\ O_2''=\mathrm{diag}\left\{O_2,\frac{1}{3}O_2\right\}$$

For $\dot{V}_7(t)$, an upper bound is given by

$$\begin{aligned}\dot{V}_7(t) &\leqslant e^{2at}\Big(l^2F^*(\hbar(t))W_1F(\hbar(t)) \\ &\quad - le^{-2al}\int_{t-l}^{t}F^*(\hbar(s))W_1F(\hbar(s))ds \\ &\quad + \omega_2^*(t)W_2\omega_2(t) - e^{-2al}\omega_2^{l*}(t)W_2\omega_2^l(t)\Big) \\ &\leqslant e^{2at}\Big(l^2F^*(\hbar(t))W_1F(\hbar(t)) - \\ &e^{-2al}\left(\int_{t-l(t)}^{t}F(\hbar(s))ds\right)^*W_1\left(\int_{t-l(t)}^{t}F^*(\hbar(s))ds\right) \\ &\quad + \omega_2^*(t)W_2\omega_2(t) - e^{-2al}\omega_2^{l*}(t)W_2\omega_2^l(t)\Big) \end{aligned} \tag{20}$$

Applying Assumption 1 yields

$$\chi(t) = \begin{bmatrix}\phi_1(t)\\ \phi_2(t)\end{bmatrix}^*\begin{bmatrix}-\emptyset_1 & 0\\ 0 & \Upsilon\emptyset_2\Upsilon\end{bmatrix}\begin{bmatrix}\phi_1(t)\\ \phi_2(t)\end{bmatrix} \geqslant 0 \tag{21}$$

From system (7), for arbitrary invertible complex matrices G_1, G_2, G_3, G_4, by defining $\mathbb{J}(t) = \hbar^*(t)G_1 + \dot{\hbar}^*(t)G_2 + \ddot{\hbar}^*(t)G_3 + h^*(t_k)G_4$, we have the following equation

$$\begin{aligned}&2e^{2at}\mathbb{J}(t)\big(-\ddot{\hbar}(t) - \widetilde{A}(t)\dot{\hbar}(t) - \widetilde{B}(t)\hbar(t) + \widetilde{C}(t)F(\hbar(t)) \\ &+ \widetilde{D}(t)F\left(\hbar^\tau(t)\right) + \widetilde{E}(t)\int_{t-l(t)}^{t}F(\hbar(s))ds + \Gamma(H+ \\ &\Delta H)\hbar(t_k) + \Gamma(H_\tau + \Delta H_\tau)\hbar^\eta(t_k)\big) = 0 \end{aligned} \tag{22}$$

Making use of Lemma 3 and performing the necessary computations, we arrive at

$$\begin{aligned}&2\mathbb{J}(t)\Big(-\Delta A(t)\dot{\hbar}(t) - \Delta B(t)\hbar(t) + \Delta C(t)F(\hbar(t)) \\ &+\Delta D(t)F\left(\hbar^\tau(t)\right) + \Delta E(t)\int_{t-l}^{t}F(\hbar(s))ds\Big) \\ &\leqslant \varepsilon_1^{-1}\mathbb{J}(t)\mathfrak{M}_1\mathfrak{M}_1^*\mathbb{J}^*(t) + \varepsilon_2^{-1}\mathbb{J}(t)\mathfrak{M}_2\mathfrak{M}_2^*\mathbb{J}^*(t) \\ &+ \varepsilon_1\left(\mathfrak{N}_1\dot{\hbar}(t) + \mathfrak{N}_2\hbar(t)\right)^*\left(\mathfrak{N}_1\dot{\hbar}(t) + \mathfrak{N}_2\hbar(t)\right) + \\ &\varepsilon_2\left(\mathfrak{N}_3F(\hbar(t)) + \mathfrak{N}_4F\left(\hbar^\tau(t)\right) + \mathfrak{N}_5\int_{t-l}^{t}F(\hbar(s))ds\right)^* \\ &\left(\mathfrak{N}_3F(\hbar(t)) + \mathfrak{N}_4F\left(\hbar^\tau(t)\right) + \mathfrak{N}_5\int_{t-l}^{t}F(\hbar(s))ds\right) \end{aligned} \tag{23}$$

$$\begin{aligned}&2\mathbb{J}(t)\Gamma\left(\Delta H\hbar(t_k) + \Delta H_\tau\hbar^\eta(t_k)\right) \\ &\leqslant \mu^{-1}\mathbb{J}(t)\Gamma KK^*\Gamma^*\mathbb{J}^*(t) \\ &+ \mu\left(N\hbar(t_k) + N_\tau\hbar^\eta(t_k)\right)^*\left(N\hbar(t_k) + N_\tau\hbar^\eta(t_k)\right) \end{aligned} \tag{24}$$

Combining the above formulas (14)-(24) results in

$$\begin{aligned}\dot{V}(t) &\leqslant \sum_{j=1}^{7}\dot{V}_j(t) + e^{2at}\chi(t) \\ &\leqslant e^{2at}\zeta^*(t)\Omega\zeta(t) \end{aligned} \tag{25}$$

where

$$\begin{aligned}\Omega &= \frac{d_2(t)}{h_k}(\Omega_1 + h_k\Omega_2) + \frac{d_1(t)}{h_k}(\Omega_1 + h_k\Omega_3) \\ \Omega_1 &= \Omega_{11} - \mathcal{T}_1 \circledast_1^{-1}\mathcal{T}_1^* \\ \Omega_2 &= \Omega_{22} + \frac{1}{h_k}e^{-2ah}\Pi_1^*Y_1O_2^{''-1}Y_1^*\Pi_1 \\ \Omega_3 &= \Omega_{33} + \frac{1}{h_k}e^{-2ah}\Pi_2^*Y_2O_1^{''-1}Y_2^*\Pi_2 \end{aligned}$$

The validity of the inequality $\Omega < 0$ is guaranteed by the Schur complement when conditions (9)-(12) hold. Thus, we have

$$\dot{V}(t) < 0, t \in \left[t_k, t_{k+1}\right) \tag{26}$$

For $t \in \left[t_k, t_{k+1}\right)$, it follows from (26) that

$$V(t) \leqslant V(t_k) \leqslant V(t_{k-1}) \leqslant \ldots \leqslant V(0) \tag{27}$$

It is noted that $V_2(0) = V_3(0) = V_6(0) = 0$. For $t \in \left[t_k, t_{k+1}\right)$, based on (13), we can deduce that

$$V(0) \leqslant \pounds\left(\sup_{s\in[-r,0]}\left\{\|\hbar(s)\|, \|\dot{\hbar}(s)\|\right\}\right)^2 \tag{28}$$

where $\pounds = 5\lambda_{\max}(B) + \tau_{12}^3\lambda_{\max}(M_1) + 2\tau_2\lambda_{\max}(M_2) + \tau_1^3\lambda_{\max}(M_3) + 2\eta\lambda_{\max}(N_1) + \eta^3\lambda_{\max}(N_2) + l^3\Upsilon^2\lambda_{\max}(W_1) + 2l\lambda_{\max}(W_2)$.

For $t \in \left[t_k, t_{k+1}\right)$, integrating system (7) over $[t_k, t]$ and applying the Lemma 7 gives

$$\begin{aligned}&\|\dot{\hbar}(t)\|^2 \leqslant 15\|\dot{\hbar}(t_k)\|^2 + 15h\Big(\int_{t_k}^{t}\bar{A}\|\dot{\hbar}(\theta)\|^2d\theta \\ &\int_{t_k}^{t}\bar{B}\|\hbar(\theta)\|^2d\theta + \int_{t_k}^{t}\bar{C}\|F(\hbar(\theta))\|^2d\theta \\ &+ \int_{t_k}^{t}\bar{D}\|F\left(\hbar^\tau(\theta)\right)\|^2d\theta + \int_{t_k}^{t}\bar{E}\|\int_{\theta-l}^{\theta}F(\hbar(s))ds\|^2d\theta \\ &+ \int_{t_k}^{t}\|\Gamma\|^2\bar{H}\|\hbar(t_k)\|^2d\theta + \int_{t_k}^{t}\|\Gamma\|^2\bar{H}_\tau\|\hbar^\eta(t_k)\|^2d\theta\Big) \end{aligned}$$

$$\leqslant 15\|\dot{\hbar}(t_k)\|^2 + 15h\bar{A}\int_{t_k}^{t}\|\dot{\hbar}(\theta)\|^2 d\theta$$
$$+ 15h\left(\bar{B} + \bar{C}\|\Upsilon\|^2 + \frac{\bar{D}}{1-d}\|\Upsilon\|^2\right)\int_{t_k}^{t}\|\hbar(\theta)\|^2 d\theta$$
$$+ 15h\frac{\bar{D}}{1-d}\|\Upsilon\|^2\int_{t_k-\tau_2}^{t}\|\hbar(\theta)\|^2 d\theta$$
$$+ 15h^2\|\Gamma\|^2\bar{H}\|\hbar(t_k)\|^2 + 15h^2\|\Gamma\|^2\bar{H}_\tau\|\hbar^\eta(t_k)\|^2$$
$$+ 15h\bar{E}\int_{t_k}^{t}\|\int_{\theta-l}^{\theta}F(\hbar(s))ds\|^2 d\theta \tag{29}$$

where

$$\bar{\mho} = \|\mho\|^2 + \mathfrak{b},\ \mho = A, B, C, D, E, H, H_\tau$$
$$\mathfrak{b} = \max\{\|\mathfrak{M}_1\|^2\|\mathfrak{N}_1\|^2, \|\mathfrak{M}_1\|^2\|\mathfrak{N}_2\|^2, \|\mathfrak{M}_2\|^2\|\mathfrak{N}_3\|^2, \|\mathfrak{M}_2\|^2\|\mathfrak{N}_4\|^2, \|\mathfrak{M}_2\|^2\|\mathfrak{N}_5\|^2\}$$

Note that the equality $\hbar(t) = \hbar(t_k) + \int_{t_k}^{t}\dot{\hbar}(s)ds$ holds, one obtains

$$\|\hbar(t)\|^2 \leqslant 2\|\hbar(t_k)\|^2 + 2h_k\int_{t_k}^{t}\|\dot{\hbar}(s)\|^2 ds \tag{30}$$

By combining (29) with (30), we have

$$\|\hbar(t)\|^2 + \|\dot{\hbar}(t)\|^2 + \|\int_{t-l}^{t}F(\hbar(s))ds\|^2$$
$$\leqslant \max\left\{15, 15h^2\|\Gamma\|^2\bar{H} + 2\right\}\left(\|\dot{\hbar}(t_k)\|^2 + \|\hbar(t_k)\|^2\right)$$
$$+ \|\Upsilon\|^2\int_{t_k-l}^{t_{k+1}}\|\hbar(s)\|^2 ds + 15h^2\|\Gamma\|^2\bar{H}_\tau\|\hbar^\eta(t_k)\|^2$$
$$+ \beta\int_{t_k}^{t}\left(\|\hbar(\theta)\|^2 + \|\dot{\hbar}(\theta)\|^2 + \|\int_{\theta-l}^{\theta}F(\hbar(s))ds\|^2\right)d\theta$$
$$+ 15h\frac{\bar{D}}{1-d}\|\Upsilon\|^2\int_{t_k-\tau_2}^{t_k}\|\hbar(s)\|^2 ds \tag{31}$$

Based on Lemma 5, one can get

$$\|\hbar(t)\|^2 + \|\dot{\hbar}(t)\|^2 + \|\int_{t-l}^{t}F(\hbar(s))ds\|^2$$
$$\leqslant \max\left\{15, 15h^2\|\Gamma\|^2\bar{H} + 2\right\}e^{h_k\beta}\times$$
$$\left(\|\dot{\hbar}(t_k)\|^2 + \|\hbar(t_k)\|^2\right) + 15h^2\|\Gamma\|^2\bar{H}_\tau e^{h_k\beta}\|\hbar^\eta(t_k)\|^2$$
$$+ \|\Upsilon\|^2 e^{h_k\beta}\int_{t_k-l}^{t_{k+1}}\|\hbar(s)\|^2 ds$$
$$+ 15h\frac{\bar{D}}{1-d}\|\Upsilon\|^2 e^{h_k\beta}\int_{t_k-\tau_2}^{t_k}\|\hbar(s)\|^2 ds \tag{32}$$

where

$$\beta = \max\left\{15h\left(\bar{B} + \bar{C}\|\Upsilon\|^2 + \frac{2\bar{D}}{1-d}\|\Upsilon\|^2\right),\ 15h\bar{A} + 2h, 15h\bar{E}\right\}$$

Form (13) and (32), it can be inferred that

$$V(t_k) \geqslant e^{2at_k}\lambda_{\min}(B)\left(\|\dot{\hbar}(t_k)\|^2 + \|\hbar(t_k)\|^2\right)$$
$$V(t_k) \geqslant e^{2a(t_k-\tau_2)}\lambda_{\min}(M_2)\int_{t_k-\tau_2}^{t_k}\|\hbar(s)\|^2 ds$$
$$V(t_k) \geqslant e^{2at_k}\lambda_{\min}(B)\|\hbar^\eta(t_k)\|^2$$
$$V(t_k) \geqslant e^{2a(t_k-l)}\lambda_{\min}(W_2)\int_{t_k-l}^{t_k}\|\hbar(s)\|^2 ds$$
$$V(t_{k+1}) \geqslant e^{2at_k}\lambda_{\min}(X)\int_{t_k}^{t_{k+1}}\|\hbar(s)\|^2 ds \tag{33}$$

Substituting (27), (28), and (33) into (32), we have

$$\|\hbar(t)\|^2 + \|\hbar(t)\|^2 + \|\int_{t-l}^{t}F(\hbar(s))ds\|$$
$$\leqslant \psi e^{-2at_k}V(t_k) + \frac{\|\Upsilon\|^2 e^{h_k\beta}}{\lambda_{\min}(X)}e^{-2at_k}V(t_{k+1})$$
$$\leqslant \psi e^{2ah_k}V(0)e^{-2at} + \frac{\|\Upsilon\|^2 e^{h_k\beta}}{\lambda_{\min}(X)}e^{2ah_k}V(0)e^{-2at}$$
$$\leqslant \mathscr{A}\pounds e^{-2at}\left(\sup_{s\in[-r,0]}\{\|\hbar(s)\|, \|\hbar(s)\|\}\right)^2 \tag{34}$$

where

$$\psi = \frac{e^{h_k\beta}}{\lambda_{\min}(B)}\Big(\max\left\{15, 15h^2\|\Gamma\|^2\bar{H} + 2\right\}$$
$$+ 15h^2\|\Gamma\|^2\bar{H}_\tau + 15h\frac{e^{2a\tau_2}\lambda_{\min}(B)}{\lambda_{\min}(M_2)}\cdot\frac{\bar{D}^2}{1-d}\|\Upsilon\|^2$$
$$\frac{e^{2al}\lambda_{\min}(B)}{\lambda_{\min}(W_2)}\|\Upsilon\|^2\Big)$$
$$\mathscr{A} = \psi e^{2ah_k} + \frac{\|\Upsilon\|^2 e^{h_k\beta}}{\lambda_{\min}(X)}e^{2ah_k}$$

So, one finally gets

$$\|\hbar(t)\| \leqslant e^{-at}\sqrt{\mathscr{A}\pounds}\sup_{s\in[-r,0]}\left\{\|\hbar(s)\|, \|\dot{\hbar}(s)\|\right\} \tag{35}$$

Definition 1 guarantees the exponential stability of system (7), proving exponential synchronization of the drive-response systems (1) and (3). The proof is complete. □

Theorem 2 *Suppose Assumption 1 hold. For given positive scalars $h, a, \varepsilon_1, \varepsilon_2, \mu, r, \rho_1, \rho_2, \rho_3$ and ρ_4, under the controller (5), system (4) is exponentially stabilized, if there exist positive definite Hermitian matrices $\tilde{B}$, $\tilde{M}_1$, $\tilde{M}_2$, $\tilde{M}_3$, $\tilde{N}_1$, $\tilde{N}_2$, $\tilde{O}_1$, $\tilde{O}_2$, $\tilde{W}_1$, $\tilde{W}_2$, $\tilde{X}$, Hermitian matrix $\tilde{Y}$, arbitrary complex matrices $\tilde{S}_1$, $\tilde{S}_2$, $\tilde{S}_3$, $\tilde{Y}_1$, $\tilde{Y}_2$, G, $\tilde{\mathcal{K}}$, R, R_τ, and real positive diagonal matrices $\tilde{\emptyset}_1$, $\tilde{\emptyset}_2$ such that*

$$\begin{bmatrix} \tilde{M}_1 & \tilde{\mathcal{K}} \\ * & \tilde{M}_1 \end{bmatrix} > 0 \tag{36}$$

$$\tilde{\Omega}_{11} - \tilde{\mathcal{T}}_1 \circledast_1^{-1} \tilde{\mathcal{T}}_1^* < 0 \tag{37}$$

$$\tilde{\Omega}_{11} + h\tilde{\Omega}_{22} - \tilde{\mathcal{T}}_2 \tilde{\circledast}_2^{-1} \tilde{\mathcal{T}}_2^* < 0 \tag{38}$$

$$\tilde{\Omega}_{11} + h\tilde{\Omega}_{33} - \tilde{\mathcal{T}}_3 \tilde{\circledast}_3^{-1} \tilde{\mathcal{T}}_3^* < 0 \tag{39}$$

where

$$\begin{aligned}
&\tilde{\circledast}_2 = [\circledast_1, -\exp(2ah)\,\tilde{\mathcal{O}}_2''], \quad \tilde{\circledast}_3 = [\circledast_1, -\exp(2ah)\,\tilde{\mathcal{O}}_1''] \\
&\tilde{\mathcal{T}}_1 = \big[\mathbb{J}_0 \Gamma K, \mathbb{J}_0 \mathfrak{M}_1, \mathbb{J}_0 \mathfrak{M}_2, \varepsilon_1 \left(\mathfrak{N}_1 G^* u_2 + \mathfrak{N}_2 G^* u_1\right)^*, \\
&\qquad \varepsilon_2 (\mathfrak{N}_3 G^* u_{13} + \mathfrak{N}_4 G^* u_{14} + \mathfrak{N}_5 G^* u_{20})^*, \\
&\qquad \mu \left(N G^* u_9 + N_\tau G^* u_{11}\right)^* \big] \\
&\tilde{\mathcal{T}}_2 = [\tilde{\mathcal{T}}_1, \Pi_1^* \tilde{Y}_1], \quad \tilde{\mathcal{T}}_3 = [\tilde{\mathcal{T}}_1, \Pi_2^* \tilde{Y}_2^*] \\
&\tilde{\Omega}_{11} = sym \Big\{ \Phi_1^{1*} \tilde{B} \Phi_1^2 + 2a \tilde{\Omega}_1^{1*} \tilde{S}_3 \tilde{\Omega}_2^1 + \tilde{\Omega}_1^{2*} \tilde{S}_3 \tilde{\Omega}_2^1 \\
&\quad + \tilde{\Omega}_1^{1*} \tilde{S}_3 \tilde{\Omega}_2^2 + \tilde{\Omega}_3^{5*} \left(\tilde{S}_1 \tilde{\Omega}_4^1 + \tilde{S}_2 \tilde{\Omega}_5\right) \Big\} + 2a \Phi_1^{1*} \tilde{B} \Phi_1^1 \\
&\quad + \tau_1^2 u_2^* \tilde{M}_3 u_2 + \Phi_2^{1*} \tilde{M}_2 \Phi_2^1 + \tau_{12}^2 u_2^* \tilde{M}_1 u_2 \\
&\quad - e^{-2a\tau_2} \Phi_2^{2*} \tilde{M}_2 \Phi_2^2 - e^{-2a\tau_1} w_1^* \tilde{M}_3 w_1 \\
&\quad - 3e^{-2a\tau_1} w_2^* \tilde{M}_3 w_2 + e^{-2a\tau_2} w_3^* \tilde{Q} w_3 \\
&\quad + 2ah^2 \tilde{\Omega}_6^{1*} \tilde{Y} \tilde{\Omega} 6^1 + \Phi_2^{1*} \tilde{N}_1 \Phi_2^1 - e^{-2a\eta} \Phi_2^{3*} \tilde{N}_2 \Phi_2^3 \\
&\quad + \eta^2 u_2^* \tilde{N}_2 u_2 - e^{-2a\eta} \varnothing_1^* \tilde{N}_2 \varnothing_1 + u_1^* \tilde{X} u_1 \\
&\quad - 3e^{-2a\eta} \varnothing_2^* \tilde{N}_2 \varnothing_2 - e^{-2ah} \Pi_1^* \tilde{O}_1' \Pi_1 \\
&\quad - e^{-2ah} \Pi_2^* \tilde{O}_2' \Pi_2 + l^2 u_{13}^* \tilde{W}_1 u_{13} - e^{-2al} u_{20}^* \tilde{W}_1 u_{20} \\
&\quad + \Phi_2^{1*} \tilde{W}_2 \Phi_2^1 - e^{-2al} \Phi_2^{5*} \tilde{W}_2 \Phi_2^5 + \mathbb{J} \\
&\quad - u_{15}^* \tilde{\emptyset}_1 u_{15} + u_1^* \Upsilon \tilde{\emptyset}_1 \Upsilon u_{15} - u_{16}^* \tilde{\emptyset}_2 u_{16} + u_4^* \Upsilon \tilde{\emptyset}_2 \Upsilon u_4 \\
&\tilde{\Omega}_{22} = sym \Big\{ 2a \tilde{\Omega}_3^{1*} \left(\tilde{S}_1 \tilde{\Omega}_4^1 + \tilde{S}_2 \tilde{\Omega}_5\right) - \frac{1}{h_k} e^{-2ah} \Pi_1^* \tilde{Y}_2 \Pi_3 \\
&\quad + \tilde{\Omega}_3^{3*} \left(\tilde{S}_1 \tilde{\Omega}_4^1 + \tilde{S}_2 \tilde{\Omega}_5\right) + \tilde{\Omega}_3^{1*} \tilde{S}_1 \tilde{\Omega}_4^2 + \tilde{\Omega}_6^{2*} \tilde{Y} \tilde{\Omega}_6^1 \Big\} \\
&\quad + \tilde{\Omega}_6^{1*} \tilde{Y} \tilde{\Omega}_6^1 + h_k u_2^* \tilde{O}_1 u_2 - \frac{1}{h_k} e^{-2ah} \Pi_1^* \tilde{O}_1' \Pi_1 \\
&\tilde{\Omega}_{33} = sym \Big\{ 2a \tilde{\Omega}_3^{2*} \left(\tilde{S}_1 \tilde{\Omega}_4^1 + \tilde{S}_2 \tilde{\Omega}_5\right) - \frac{1}{h_k} e^{-2ah} \Pi_1^* \tilde{Y}_1 \Pi_3 \\
&\quad + \tilde{\Omega}_3^{4*} \left(\tilde{S}_1 \tilde{\Omega}_4^1 + \tilde{S}_2 \tilde{\Omega}_5\right) + \tilde{\Omega}_3^{2*} \tilde{S}_1 \tilde{\Omega}_4^2 + \tilde{\Omega}_6^{3*} \tilde{Y} \tilde{\Omega}_6^1 \Big\} \\
&\quad - \tilde{\Omega}_6^{1*} \tilde{Y} \tilde{\Omega}_6^1 + h_k u_2^* \tilde{O}_2 u_2 - \frac{1}{h_k} e^{-2ah} \Pi_2^* \tilde{O}_2' \Pi_2 \\
&\mathbb{J} = sym \big\{ \mathbb{J}_0 \big(-G^* u_3 - A G^* u_2 - B G^* u_1 + C G^* u_{13} \\
&\quad + D G^* u_{14} + E G^* u_{20} + \Gamma R u_9 + \Gamma R_\tau u_{11}\big) \big\}
\end{aligned}$$

And the controller gains $H = R(G^*)^{-1}$ and $H_\tau = R_\tau (G^*)^{-1}$.

Proof In Theorem 1, let $G_1 = \rho_1 G^{-1}$, $G_2 = \rho_2 G^{-1}$, $G_3 = \rho_3 G^{-1}$, and $G_4 = \rho_4 G^{-1}$. Defining

$$\begin{aligned}
&\mathcal{G}_j = \mathrm{diag}\{\underbrace{G, \cdots, G}_{j}\}, \ j = 2, 4, 6 \\
&\tilde{B} = \mathcal{G}_4 B \mathcal{G}_4^*, \ \tilde{Y} = \mathcal{G}_6 Y \mathcal{G}_6^*, \ \tilde{S}_1 = \mathcal{G}_4 S_1 \mathcal{G}_4^*, \ \tilde{S}_2 = \mathcal{G}_4 S_2 \mathcal{G}_4^* \\
&\tilde{S}_3 = \mathcal{G}_2 \tilde{S}_3 \mathcal{G}_2^*, \ \tilde{Y}_1 = \mathcal{G}_2 Y_1 \mathcal{G}_2^*, \ \tilde{Y}_2 = \mathcal{G}_2 Y_2 \mathcal{G}_2^*, \ \tilde{N}_1 = \mathcal{G}_2 N_1 \mathcal{G}_2^* \\
&\tilde{N}_2 = G N_2 G^*, \ \tilde{\mathcal{K}} = G \mathcal{K} G^*, \ \tilde{\emptyset}_1 = G \emptyset_1 G^*, \ \tilde{\emptyset}_2 = G \emptyset_2 G^* \\
&\tilde{W}_1 = G W_1 G^*, \ \tilde{W}_2 = \mathcal{G}_2 W_2 \mathcal{G}_2^*, \ \tilde{M}_1 = G M_1 G^* \\
&\tilde{M}_2 = G M_2 G^*, \ \tilde{M}_3 = G M_3 G^*, \ \tilde{X} = G X G^* \\
&\tilde{O}_1 = G O_1 G^*, \ \tilde{O}_2 = G O_2 G^*, \ R = H G^*, \ R_\tau = H_\tau G^* \\
&\mathfrak{C}_1 = \mathrm{diag}\{\underbrace{G, G, \cdots, G, I}_{22}\}, \ \mathfrak{C}_2 = \mathrm{diag}\{\mathfrak{C}_1, G, G\}.
\end{aligned}$$

Via pre-and post-multiplying both sides of (9) with $\mathcal{G}_2$ and its conjugate transpose, respectively, (36) can be obtained. Via pre-and post-multiplying both sides of (10) with $\mathfrak{C}_1$ and its conjugate transpose, respectively, (37) can be derived. Finally, via pre-and post-multiplying both sides of (11) and (12) with $\mathfrak{C}_2$ and its conjugate transpose, respectively, (38) and (39) can be obtained. This completes the proof. □

Remark 3 Theorems 1 and 2 solved the exponential synchronization of CVINNs with time-varying uncertainties under non-fragile memory SDC scheme. When considering conventional CVINNs model without parameter uncertainties, similar results can be derived to address the exponential synchronization.

For systems (4)and (7), when $\Delta A = 0, \Delta B = 0, \Delta C = 0, \Delta D = 0, \Delta E = 0$, the nominal systems are written as

$$\begin{aligned}
\ddot{\hbar}(t) = &-A\dot{\hbar}(t) - B\hbar(t) + CF(\hbar(t)) + DF\left(\hbar^\tau(t)\right) \\
&+ E\int_{t-l(t)}^{t} F(\hbar(s))ds
\end{aligned} \tag{40}$$

$$\ddot{\hbar}(t) = -A\dot{\hbar}(t) - B\hbar(t) + CF(\hbar(t)) + DF\left(\hbar^\tau(t)\right)$$

$$+ E\int_{t-l(t)}^{t} F(\hbar(s))ds + \Gamma(H+\Delta H)\hbar(t_k) + \Gamma(H_\tau+\Delta H_\tau)\hbar^{\eta}(t_k) \tag{41}$$

Corollary 1 *For system (41), under Assumption 1, and given positive scalars h, a, r, μ, it is exponentially stable, if there exist positive definite Hermitian matrices B, M_1, M_2, M_3, N_1, N_2, O_1, O_2, W_1, W_2, X, Hermitian matrix Y, arbitrary complex matrices S_1, S_2, S_3, Y_1, Y_2, G_1, G_2, G_3, G_4, $\mathcal{K}$, and real positive diagonal matrices $\emptyset_1, \emptyset_2$ such that*

$$\begin{bmatrix} M_1 & \mathcal{K} \\ * & M_1 \end{bmatrix} > 0 \tag{42}$$

$$\Omega_{11} - \bar{\mathcal{T}}_1 \bar{\circledast}_1^{-1} \bar{\mathcal{T}}_1^* < 0 \tag{43}$$

$$\Omega_{11} + h\Omega_{22} - \bar{\mathcal{T}}_2 \bar{\circledast}_2^{-1} \bar{\mathcal{T}}_2^* < 0 \tag{44}$$

$$\Omega_{11} + h\Omega_{33} - \bar{\mathcal{T}}_3 \bar{\circledast}_3^{-1} \bar{\mathcal{T}}_3^* < 0 \tag{45}$$

where

$$\bar{\circledast}_1 = [-\mu I, -\mu I],\ \bar{\circledast}_2 = [\bar{\circledast}_1, -\exp(2ah)\,\mathcal{O}_2^{''}]$$
$$\bar{\circledast}_3 = [\bar{\circledast}_1, -\exp(2ah)\,\mathcal{O}_1^{''}]$$
$$\bar{\mathcal{T}}_1 = [\mathbb{J}_0 \Gamma K, \mu(Nu_9 + N_\tau u_{11})^*]$$
$$\bar{\mathcal{T}}_2 = [\bar{\mathcal{T}}_1, \Pi_1^* Y_1],\ \bar{\mathcal{T}}_3 = [\bar{\mathcal{T}}_1, \Pi_2^* Y_2^*]$$

and other notations are the same as those in Theorem 1.

Corollary 2 *For system (40), under Assumption 1, and given positive scalars h, a, r, μ, $\rho_1, \rho_2, \rho_3, \rho_4$, it is exponentially stabilized via the controller (5), if there exist positive definite Hermitian matrices $\tilde{B}$, $\tilde{M}_1, \tilde{M}_2, \tilde{M}_3$, $\tilde{N}_1, \tilde{N}_2$, $\tilde{O}_1, \tilde{O}_2$, $\tilde{W}_1, \tilde{W}_2$, $\tilde{X}$, Hermitian matrix $\tilde{Y}$, arbitrary complex matrices $\tilde{S}_1, \tilde{S}_2, \tilde{S}_3$, $\tilde{Y}_1, \tilde{Y}_2$, G, $\tilde{\mathcal{K}}$, R, R_τ, and real positive diagonal matrices $\tilde{\emptyset}_1, \tilde{\emptyset}_2$ such that*

$$\begin{bmatrix} \tilde{M}_1 & \tilde{\mathcal{K}} \\ * & \tilde{M}_1 \end{bmatrix} > 0 \tag{46}$$

$$\tilde{\Omega}_{11} - \widehat{\mathcal{T}}_1 \widehat{\circledast}_1^{-1} \widehat{\mathcal{T}}_1^* < 0 \tag{47}$$

$$\tilde{\Omega}_{11} + h\tilde{\Omega}_{22} - \widehat{\mathcal{T}}_2 \widehat{\circledast}_2^{-1} \widehat{\mathcal{T}}_2^* < 0 \tag{48}$$

$$\tilde{\Omega}_{11} + h\tilde{\Omega}_{33} - \widehat{\mathcal{T}}_3 \widehat{\circledast}_3^{-1} \widehat{\mathcal{T}}_3^* < 0 \tag{49}$$

where

$$\widehat{\circledast}_1 = [-\mu I, -\mu I],\ \widehat{\circledast}_2 = [\widehat{\circledast}_1, -\exp(2ah)\,\tilde{\mathcal{O}}_2^{''}]$$
$$\widehat{\circledast}_3 = [\widehat{\circledast}_1, -\exp(2ah)\,\tilde{\mathcal{O}}_1^{''}]$$
$$\widehat{\mathcal{T}}_1 = [\mathbb{J}_0 \Gamma K, \mu\left(NG^* u_9 + N_\tau G^* u_{11}\right)^*]$$
$$\widehat{\mathcal{T}}_2 = [\tilde{\mathcal{T}}_1, \Pi_1^* \tilde{Y}_1],\ \widehat{\mathcal{T}}_3 = [\tilde{\mathcal{T}}_1, \Pi_2^* \tilde{Y}_2^*]$$

and other notations are the same as those in Theorem 2.

Remark 4 Distinct from [13–15], which focus exclusively on real-valued INNs, our research extends the domain to complex one while simultaneously accounting for the influence of mixed delays. By constructing Lyapunov functionals and employing sophisticated inequality techniques, the requisite results are obtained. To the extent of our knowledge, this constitutes the first comprehensive examination of exponential synchronization for uncertain CVINNs with mixed delays within the framework of a memory non-fragile sampled-data scheme.

Remark 5 Predominantly, existing studies concerning CVINNs [10,32] advocate order reduction via variable substitution, which not only augments the quantity of state variables and system dimensions but also complicates computational and analytical processes, neglecting the impact of inertia properties on system dynamics. To circumvent these limitations, we opt for a direct analysis of the original second-order system, thereby preserving its intrinsic characteristics and ensuring greater relevance to practical applications.

4 Illustrative examples

The efficacy of the established conclusions will be illustrated through a numerical example and an application.

Example 1 Consider systems (1) and (3) with the following parameters:

$$A = \begin{bmatrix} 0.9 & 0 \\ 0 & 0.5 \end{bmatrix},\ B = \begin{bmatrix} 0.3 & 0 \\ 0 & 0.3 \end{bmatrix}$$
$$C = \begin{bmatrix} -0.5+1i & 0.6+0.2i \\ -1.5-1i & 0.4+2.5i \end{bmatrix}$$
$$D = \begin{bmatrix} -1.5+0.1i & -1+2.1i \\ 3.1+1.1i & -1.1-1i \end{bmatrix}$$

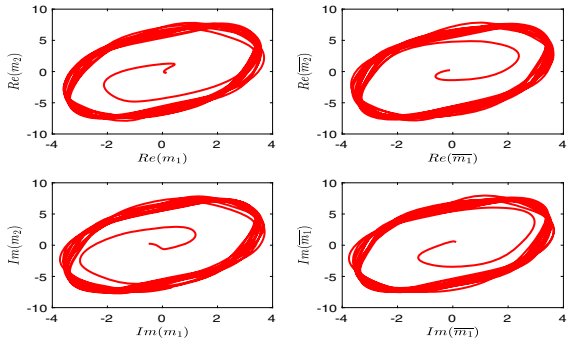

Fig. 1 Chaotic behaviors of system (1) and (3)

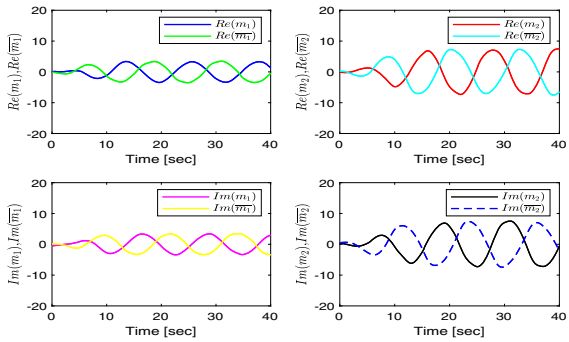

Fig. 2 Trajectories of states without the controller

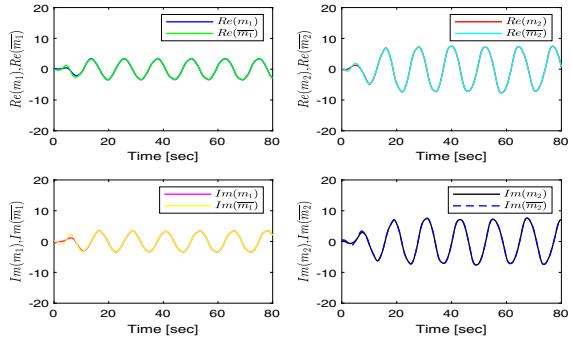

Fig. 3 Trajectories of states with the controller

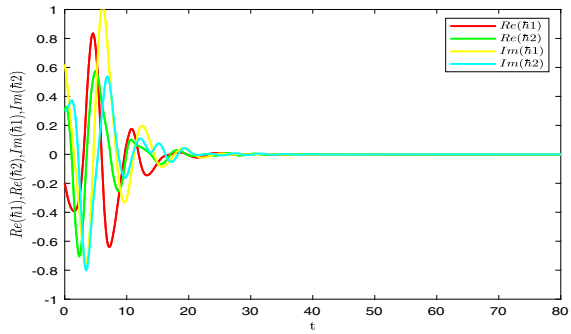

Fig. 4 Curves of error state under controller

$$E = \begin{bmatrix} -0.1+1i & 0.35-1i \\ 0.3+1.1i & 1.1-1i \end{bmatrix}, \Gamma = \begin{bmatrix} 0.12 & 0.1 \\ 0.2 & -0.1 \end{bmatrix}$$

$$\mathfrak{M}_1 = \mathfrak{M}_2 = I_2, \mathfrak{N}_j = 0.2I_2 (j = 1, 2, ..., 5)$$

$$S(t) = \sin(2t), K = 0.1I_2, N = N_\tau = 0.5I_2$$

$$\Theta(t) = \sin(t), (t) = 0.5\sin^2(t)$$

$$\tau(t) = 0.1\sin^2(t) + 0.1, \eta = 0.1$$

and complex-valued activation functions are selected as $f(m(t)) = 0.5\tanh(m(t))$, and therefore Assumption 1 holds with $\Upsilon = \text{diag}\{0.5, 0.5\}$. Let the initial conditions be $\nu_1(s) = \text{col}\{0.1 - 0.5i, -0.1 + 0.2i\}$, $\nu_2(s) = \text{col}\{0.2 - 0.1i, -0.15 + 0.2i\}$, $\bar{\nu}_1(t) = \text{col}\{-0.1+0.12i, 0.2+0.5i\}$, $\bar{\nu}_2(t) = \text{col}\{-0.1-0.1i, 0.1+0.2i\}$. Fig. 1 shows the phase plot of the state trajectories and Fig. 2. depicts the time response of the states of drive-response systems (1) and (3) when $U(t) = 0$. We can clearly see that they are out of synchronization when $U(t) = 0$. By solving Theorem 2, we can find the ultimate maximum sampling $h = 0.10375$. Furthermore, a collection of feasible solutions to LMIs (36)-(39) can be obtained as follows:

So, under the controller (5), systems (1) and (3) exhibit exponential synchronization, with Figs. 3 and 4 presenting the state and synchronization error trajectories, respectively. The corresponding control input signals are documented in Fig. 5, while Fig. 6 characterizes the aperiodic sampling intervals. Figs. 2-6 collectively substantiate the efficacy of the proposed control framework.

Example 2 Systems (1) and (3) adopt identical parameters to Example 1. The chaotic and bounded nature of their state trajectories, shown in Fig. 1, validates the model's applicability to secure communication.

$$H = \begin{bmatrix} -0.5964 - 2.1857i & -8.1920 + 1.0798i \\ -2.4769 + 0.5250i & 3.3744 + 0.5168i \end{bmatrix}$$

and

$$H_\tau = \begin{bmatrix} -0.1075 + 0.1028i & -0.0869 - 0.0221i \\ -0.1108 - 0.0287i & 0.0438 - 0.0187i \end{bmatrix}$$

A. Encryption and Decryption steps

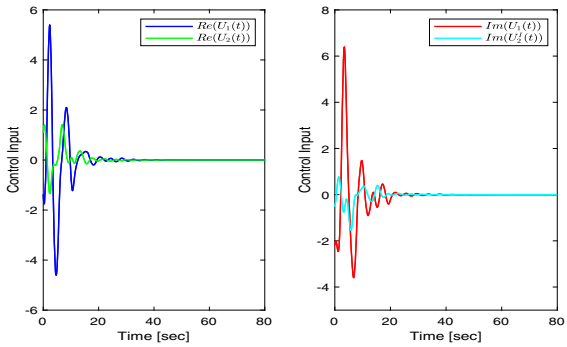

Fig. 5 Trajectories of control inputs

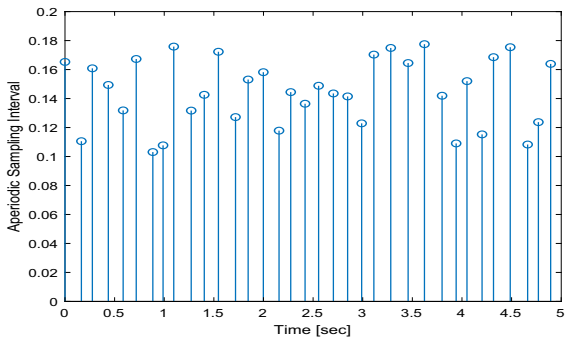

Fig. 6 Curves of aperiodic sampled-data interval

Step 1: Separate the color image Monkey of size 256×256 pixels as shown in Fig. 7(a) into three independent grayscale channels: R, G, and B.
Step 2: Ensure that the chaotic sequence generated by the main system can cover the total number of pixels $M \times N$. For any given s and k, construct the chaotic key sequences as follows: $X_1 = m_1^R(n)$, $X_2 = m_2^R(n)$, $X_3 = m_1^I(n)$, $X_4 = m_2^I(n)$, $X_5 = 0.4m_1^R(n) + 0.6m_2^I(n)$, $n = 1, 2, 3, ..., M \times N$.
Step 3: According to different rules, assign the corresponding chaotic sequences to each color channel, and use the assigned chaotic sequences to generate two key matrices: the position permutation matrix S and the XOR matrix Y. These two matrices will be used respectively for scrambling pixel positions and altering pixel values:

$$S = \mathrm{mod}\left(\left[\left(X_q + X_j\right) \times 10^4\right], M \times N\right) + 1$$
$$Y = \mathrm{mod}\left(\left[\left(X_q + X_j\right) \times 10^4\right], 256\right),$$
$$q, j = 1, 2, ..., 5, q \neq j$$

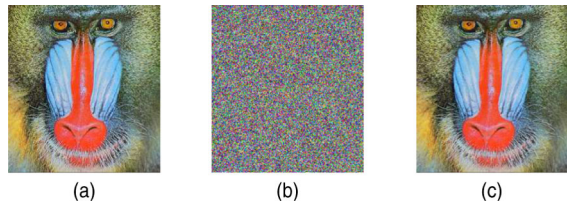

Fig. 7 (a) Color image "Houzi"; (b) The encrypted image; (c) The decrypted image

Step 4: Perform encryption operations on each color channel separately. First, flatten S into a one-dimensional vector and sort it to obtain the index mapping σ. Rearrange the channel according to σ to get the permuted channel 'channel2'. Then, perform a pixel-wise XOR operation on the permuted matrix to obtain the encrypted channel 'channel3'.

$$\mathrm{channel2}\,(q, j) = \mathrm{channel}\left(\sigma^{-1}(q, j)\right)$$
$$\mathrm{channel3} = \mathrm{bitxor}\,(\mathrm{channel2}, Y)$$

Finally, the encrypted image Fig. 7(b) is output.
Step 5: Under the controller (5), systems (1) and (3) achieve the exponential synchronization. Therefore, we select the corresponding chaotic sequences to preprocess the data in the response system, followed by performing inverse operations of step 4 on the encrypted image. Finally, the decrypted image Fig. 7(c) is output.
B. Simulation results and analysis
1) Histogram Analysis. The histograms of the original image, encrypted image, and decrypted image in RGB channels are depicted in Fig. 8, respectively. Clearly, the original image's RGB channel histograms exhibit significantly non-uniform distributions, indicating pixel value concentration in specific grayscale ranges. After encryption, all channel histograms transform into approximately uniform distributions, effectively eliminating the original image's statistical characteristics and resisting histogram-based attacks. The decrypted image's histograms overlap perfectly with the original, demonstrating the algorithm's excellent reversible recovery capability.
2) Pixel Correlation Analysis. To evaluate the encryption effect, we calculate the correlation coefficient (CC) of adjacent pixels in the images before and after encryption, including both horizontal and vertical correlations. Experimental results show that the horizontal and vertical correlation coefficients of the original image are 0.8004 and 0.7999, respectively. After encryption,

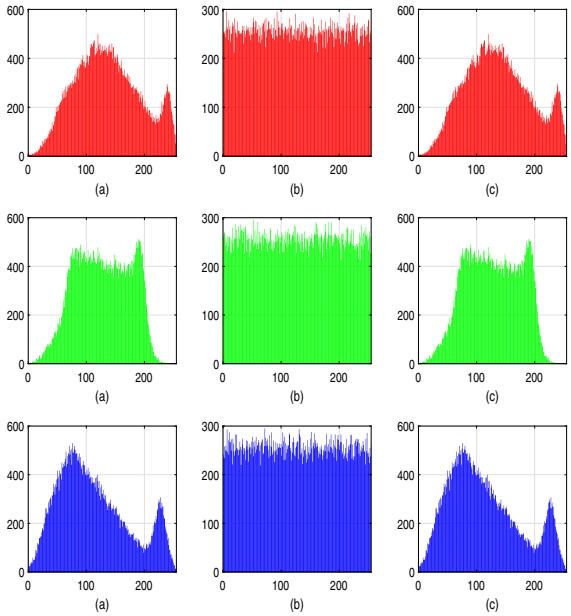

Fig. 8 Histogram of image "Houzi". The first, second, and third columns represent the histograms of the original image, encrypted image, and decrypted image in RGB channels, respectively

these values drop to 0.0469 and 0.0443, representing a reduction of over 94%. This indicates that the encryption process significantly weakened the correlation between pixels, meeting the requirements of an ideal encryption system. Here, the formula of CC is as follows

$$\mathrm{CC} = \frac{\sum_{u=1}^{m}\sum_{v=1}^{n}(s_{uv}-\bar{s})\left(k_{uv}-\bar{k}\right)}{\sqrt{\left(\sum_{u=1}^{m}\sum_{v=1}^{n}(s_{uv}-\bar{s})^2\right)\left(\sum_{u=1}^{m}\sum_{v=1}^{n}\left(k_{uv}-\bar{k}\right)^2\right)}}$$

3) Decryption Quality Assessment. The decryption accuracy is measured using Mean Absolute Error (MAE) and Peak Signal-to-Noise Ratio (PSNR). Calculations yields an MAE of 0.0007, which is close to zero, indicating minimal pixel value deviation between the decrypted image and the original image. The PSNR is 61.15, demonstrating that the algorithm completely destroys the statistical features of the image while enabling nearly lossless image recovery through precise key synchronization. Here, the formulas of MAE and PSNR are as follows

$$\mathrm{MAE} = \frac{1}{mn}\sum_{u=1}^{m}\sum_{v=1}^{n}|s(u,v)-k(u,v)|$$

$$\mathrm{PSNR} = 20\log_{10}\left[\frac{255}{\sqrt{\mathrm{MSE}}}\right]$$

$$\mathrm{MSE} = \frac{1}{mn}\sum_{u=1}^{m}\sum_{v=1}^{n}(s(u,v)-k(u,v))^2$$

These results prove the algorithm achieves near-lossless image recovery through precise key synchronization while completely disrupting statistical features.

5 Conclusion

The exponential synchronization problem of CVINNs with mixed time delays and parameter uncertainties has been investigated. To address these issues, a memory non-fragile SDC incorporating a time-delay term is proposed for the first time to ensure the stability of the closed-loop system. Additionally, the improved reciprocally convex inequality has been extended to the complex domain to be fully utilized. The synchronization criteria have been established through the construction of Lyapunov functionals combined with innovative inequality treatments, ultimately expressed as feasible LMIs. To validate the theoretical developments, numerical simulations and application case studies are systematically presented. Considering that network attacks frequently occur, it is worth further investigating the performance analysis of CVINNs under network attacks in the future.

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Declarations

Competing interests The authors declare no competing interests.

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